

CS 253 Mark Holliday (written by Dr. Andrew Scott and Dr. Mark Holliday)

Project 1: uk.json Project 1 Output

The following is the output from using the `uk.json` file. This assumes that `uk.json` is in the same directory as the client and server JavaScript files.

Section 1: Network tests

In one shell the command below is used.

```
node railway_network.js
```

In another shell the command below is used

```
node client_network.js uk.json
```

The output is below.

```
NETWORK Tests

===Network TEST=1=NETWORK=NAME===
United Kingdom Railway System

===Network TEST=2=GETTING=ROUTES=ARRAY===
There are 14 routes on this network
The type of the routes is object

===Network TEST=3=ROUTE=NAMES===
[
  'East Coast Main Line',
  'West Coast Main Line',
  'Great Western Railway Line',
  'Network South West Line',
  'South Western Main Line',
  'Cherwell Valley Line',
  'Great Eastern Main Line',
  'High Speed 1 Line',
  'Midland Main Line',
  'Wales Main Line',
  'Central Wales Line',
  'West Scotland Line',
  'Edinburgh Main Line',
  'Glasgow-Edinburgh Line'
]

===Network TEST=4=ROUTE=NAMES=TOSTRING===
```

```
East Coast Main Line,  
West Coast Main Line,  
Great Western Railway Line,  
Network South West Line,  
South Western Main Line,  
Cherwell Valley Line,  
Great Eastern Main Line,  
High Speed 1 Line,  
Midland Main Line,  
Wales Main Line,  
Central Wales Line,  
West Scotland Line,  
Edinburgh Main Line,  
Glasgow-Edinburgh Line
```

```
===Network TEST=5=Total_Stations===  
There are 56 stations in this network.
```

```
===Network TEST=6=FIND=LONGEST=ROUTE===  
Longest route is: ROUTE:West Coast Main Line(blue)  
STATIONS:  
1 London 0 miles  
2 St Albans 60 miles  
3 Northampton 130 miles  
4 Birmingham 195 miles  
5 Wrexham 265 miles  
6 Liverpool 305 miles  
7 Manchester 340 miles  
8 Lancaster 390 miles  
9 Kendall 450 miles  
10 Carlisle 525 miles  
11 Glasgow 635 miles  
Total Route Distance: 635 miles
```

Section 2: Route summary tests

In one shell the command below is used.

```
node railway_routeSummary.js
```

In another shell the command below is used

```
node client_routeSummary.js uk.json
```

The output is below.

ROUTE SUMMARY Tests

```
===Route Summary TEST=1=ROUTE=SUMMARY===
```

Routes Summary

```
=====
```

East Coast Main Line - London	to Edinburgh	- 500 miles
West Coast Main Line - London	to Glasgow	- 635 miles
Great Western Railway Line - London	to Fishguard	- 325 miles
Network South West Line - Birmingham	to Truro	- 345 miles
South Western Main Line - London	to Plymouth	- 245 miles
Cherwell Valley Line - Reading	to Birmingham	- 125 miles
Great Eastern Main Line - London	to Kings Lynn	- 170 miles
High Speed 1 Line - London	to Paris	- 295 miles
Midland Main Line - Birmingham	to York	- 240 miles
Wales Main Line - Cardiff	to Holyhead	- 235 miles
Central Wales Line - Cardiff	to Aberystwyth	- 175 miles
West Scotland Line - Stranraer	to Fort William	- 270 miles
Edinburgh Main Line - Edinburgh	to John O'Groats	- 323 miles
Glasgow-Edinburgh Line - Stranraer	to Edinburgh	- 185 miles

```
===Route Summary TEST=2=ROUTE=BY=NAME=(ASC)===
```

Routes Summary

```
=====
```

Central Wales Line - Cardiff	to Aberystwyth	- 175 miles
Cherwell Valley Line - Reading	to Birmingham	- 125 miles
East Coast Main Line - London	to Edinburgh	- 500 miles
Edinburgh Main Line - Edinburgh	to John O'Groats	- 323 miles
Glasgow-Edinburgh Line - Stranraer	to Edinburgh	- 185 miles
Great Eastern Main Line - London	to Kings Lynn	- 170 miles
Great Western Railway Line - London	to Fishguard	- 325 miles
High Speed 1 Line - London	to Paris	- 295 miles
Midland Main Line - Birmingham	to York	- 240 miles
Network South West Line - Birmingham	to Truro	- 345 miles
South Western Main Line - London	to Plymouth	- 245 miles
Wales Main Line - Cardiff	to Holyhead	- 235 miles
West Coast Main Line - London	to Glasgow	- 635 miles
West Scotland Line - Stranraer	to Fort William	- 270 miles

```
===Route Summary TEST=3=ROUTE=BY=NAME=(DESC)===
```

Routes Summary

```
=====
```

West Scotland Line - Stranraer	to Fort William	- 270 miles
West Coast Main Line - London	to Glasgow	- 635 miles
Wales Main Line - Cardiff	to Holyhead	- 235 miles
South Western Main Line - London	to Plymouth	- 245 miles
Network South West Line - Birmingham	to Truro	- 345 miles
Midland Main Line - Birmingham	to York	- 240 miles
High Speed 1 Line - London	to Paris	- 295 miles
Great Western Railway Line - London	to Fishguard	- 325 miles
Great Eastern Main Line - London	to Kings Lynn	- 170 miles
Glasgow-Edinburgh Line - Stranraer	to Edinburgh	- 185 miles
Edinburgh Main Line - Edinburgh	to John O'Groats	- 323 miles
East Coast Main Line - London	to Edinburgh	- 500 miles

```

Cherwell Valley Line - Reading          to Birmingham - 125 miles
Central Wales Line   - Cardiff          to Aberystwyth - 175 miles

===Route Summary TEST=4=ROUTE=BY=LENGTH=(ASC)===
Routes Summary
=====
Cherwell Valley Line - Reading          to Birmingham - 125 miles
Great Eastern Main Line - London        to Kings Lynn  - 170 miles
Central Wales Line   - Cardiff          to Aberystwyth - 175 miles
Glasgow-Edinburgh Line - Stranraer      to Edinburgh   - 185 miles
Wales Main Line      - Cardiff          to Holyhead    - 235 miles
Midland Main Line    - Birmingham       to York        - 240 miles
South Western Main Line - London        to Plymouth    - 245 miles
West Scotland Line   - Stranraer        to Fort William - 270 miles
High Speed 1 Line    - London           to Paris       - 295 miles
Edinburgh Main Line  - Edinburgh        to John O'Groats - 323 miles
Great Western Railway Line - London      to Fishguard   - 325 miles
Network South West Line - Birmingham    to Truro       - 345 miles
East Coast Main Line - London           to Edinburgh   - 500 miles
West Coast Main Line - London           to Glasgow     - 635 miles

===Route Summary TEST=5=ROUTE=BY=LENGTH=(DESC)===
Routes Summary
=====
West Coast Main Line - London          to Glasgow - 635 miles
East Coast Main Line - London          to Edinburgh - 500 miles
Network South West Line - Birmingham   to Truro - 345 miles
Great Western Railway Line - London     to Fishguard - 325 miles
Edinburgh Main Line - Edinburgh        to John O'Groats - 323 miles
High Speed 1 Line - London             to Paris - 295 miles
West Scotland Line - Stranraer         to Fort William - 270 miles
South Western Main Line - London        to Plymouth - 245 miles
Midland Main Line - Birmingham         to York - 240 miles
Wales Main Line - Cardiff              to Holyhead - 235 miles
Glasgow-Edinburgh Line - Stranraer     to Edinburgh - 185 miles
Central Wales Line - Cardiff           to Aberystwyth - 175 miles
Great Eastern Main Line - London        to Kings Lynn - 170 miles
Cherwell Valley Line - Reading         to Birmingham - 125 miles

```

Section 3: Route tests

In one shell the command below is used.

```
node railway_route.js
```

In another shell the command below is used

```
node client_route.js uk.json "Great Western Railway Line" "Cardiff" "Reading"
```

Great Western Railway Line is the route used in the first three tests. **Cardiff** and **Reading** are the two stops for the extra credit test. The extra credit test determines if there is any single route that contains both of those stops. If there is such a route, it returns a string specifying that route and the number of stops and distance traversed getting from the first to the second stop. Note that the route (if any) found in the extra credit test need not be the same route as the route specified as a command line argument.

In this example there is a route connecting **Cardiff** and **Reading** but it is just a coincidence that route, **Great Western Railway Line**, is also a command-line argument. The output is below.

ROUTE Tests

```
===Route TEST=1=GET=ROUTE===
```

```
Found: Great Western Railway Line
```

```
===Route TEST=2=ROUTE=TO=STRING===
```

```
ROUTE:Great Western Railway Line(Magenta)
```

```
STATIONS:
```

```
1 London 0 miles
```

```
2 Reading 35 miles
```

```
3 Bath 95 miles
```

```
4 Bristol 135 miles
```

```
5 Cardiff 180 miles
```

```
6 Swansea 235 miles
```

```
7 Pembroke 290 miles
```

```
8 Fishguard 325 miles
```

```
Total Route Distance: 325 miles
```

```
===Route TEST=3=ROUTE=DISTANCE===
```

```
Distance of Line as calculated: 325
```

```
====(OPTIONAL) Route TEST=4=BONUS1=FIND=FROM=TO===
```

```
Great Western Railway Cardiff to Reading 3 stops and 145 miles
```